



课程名称: 信号与系统 开课单位: 电子系 考试时长: 2 小时

题号								
分值								

本试卷共 6 大题, 满分 100 分 (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

Problem 1 Choose the correct answer or fill the blank ($2' \times 10 = 20'$)

- The frequency response of an accumulator is a ____ filter, while the frequency response of differentiator is a ____ filter.
(a) Low-pass (b) High-pass (c) Band-pass
- Which of the following is the property of a CTFT?
(a) Periodical (b) Aperiodical (c) Unsure
- Which of the following is the period of a DTFT?
(a) The period of the DT signal analyzed (b) 2π (c) The sampling rate f_s
- The Fourier transform of a periodical CT signal is a ____ spectrum.
(a) Continuous (b) Discrete or line (c) Depending on the value of the period
- To design a CT low-pass filter with cutoff frequency 80 Hz, we can design a DT low-pass filter with cutoff frequency ____, when the sampling rate is 2000 Hz.
(a) 1/25 Hz (b) 25 Hz (c) 2/25Hz (d) others
- For both amplitude modulation and frequency modulation, the bandwidth of the modulated signal is only dependent on the bandwidth of the message signal. This statement is ____.
(a) True (b) False (c) Unsure
- Which of the following can be used in amplitude modulation as the carrier signal?
(a) Sinusoidal signal (b) Rectangular pulse train (c) Complex exponential signal

- (d) All signals in (a), (b) and (c)
8. For signal $\sin(480\pi t)$, if we use a sampling rate 400 Hz, and then an ideal low-pass filter (with cutoff frequency 200 Hz) for signal reconstruction, which of the following would be the frequency of the reconstructed signal?
- (a) 240 Hz (b) 160 Hz (c) 200 Hz (d) 480Hz
9. Which of the following means can be used to avoid aliasing?
- (a) Use ideal low-pass filter with a larger cut-off frequency in signal reconstruction.
 (b) Use a sampling rate satisfying sampling theorem.
 (c) Use time-division multiplexing during the sampling process.
10. For pulse-amplitude modulation (PAM), which of the following means can be used to avoid inter-symbol interference?
- (a) Add a frequency modulation block after PAM.
 (b) Use a pulse with band-limited frequency response as the carrier signal.
 (c) Reduce the duration of pulse in PAM.

Problem 2 (5' × 2=10')

- 1) Determine the Fourier series coefficients of signal $x[n]$, where $x[n]$ is periodic with period 6 and $x[n] = \begin{cases} 1, n = 0, 1, 2, 3 \\ 0, n = 4, 5 \end{cases}$.
- 2) Consider a DT LTI system with impulse response $h[n] = (1/2)^{|n|}$. Determine the Fourier series coefficients of the output signal $y[n]$ for the input signal $x[n]$ in 1).

Problem 3 (3'+4'+3'=10')

A causal and stable LTI system S has the frequency response of

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega + 1}{6 - \omega^2 + 5j\omega},$$

- 1) Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S .
 2) Determine the impulse response $h(t)$ of S .
 3) What is the output of S when the input is $x(t) = e^{-4t}u(t)$?

Problem 4 (8'+5'+8'+5'+4'=30')

- 1) Use the Fourier transform analysis equation to calculate the Fourier transforms of:

(a) $(\frac{1}{2})^{n-2}u[n-2]$ (b) $(\frac{1}{4})^{|n-1|}$

- 2) Determine the Fourier transform for $-\pi \leq \omega < \pi$ of the following periodic signal:

$$\cos\left(\frac{\pi}{16}n + \frac{\pi}{4}\right)$$

- 3) Given the $x[n]$ has Fourier transform $X(e^{j\omega})$, express the Fourier transform of the following signals with the terms of $X(e^{j\omega})$.

(a) $x_1[n] = x[1 - n] + x[-1 - n]$

(b) $x_2[n] = (n - 1)^2 x[n]$

- 4) Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x[n]$ depicted in Figure 1. Please evaluate $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

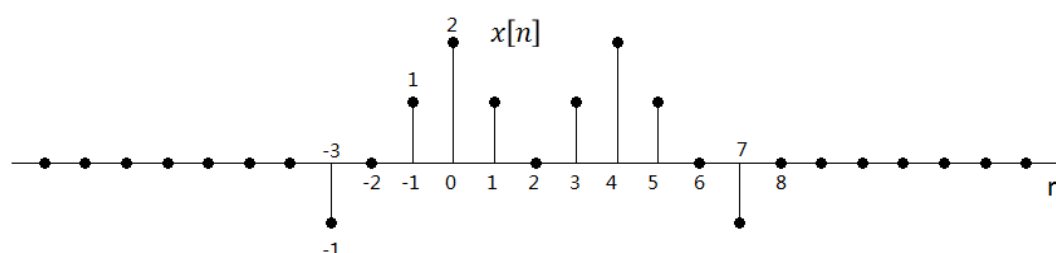


Figure 1

- 5) For a discrete-time periodic signal $x[n] = 2\cos(\pi n/6) + 4\cos(\pi n/3)$, what are the values of its Fourier series coefficients a_{2018} and a_{2019} ?

Problem 5 (12'+8'=20')

- 1) A signal $x(t)$ with Fourier transform $X(j\omega)$ undergoes impulse-train sampling to generate

$$x_p(t) = \sum_{n=-\infty}^{\infty} x[nT]\delta(t - nT),$$

where $T = 10^{-4}$. For each of the following sets of constraints on $X(j\omega)$, does the sampling theorem guarantee that $x(t)$ can be recovered exactly from $x_p(t)$?

- (a) When $|\omega| > 4000\pi$, $X(j\omega) = 0$;
 (b) When $|\omega| > 1000\pi$, $\text{Re}\{X(j\omega)\} = 0$;
 (c) $x(t)$ is real; when $\omega < -6000\pi$, $X(j\omega) = 0$.

- 2) Shown in Figure 2 is a system that processed continuous-time signals using a digital filter $h[n]$ that is linear and causal with difference equation

$$y[n] = \frac{1}{2} y[n - 1] + x[n]$$

For input signals that are band limited such that $X_c(j\omega) = 0$ for $|\omega| > \pi/T$, the system in the

figure is equivalent to a continuous-time LTI system.

Please calculate the DTFT of $h[n]$, determine the frequency response $H_c(j\omega)$ of the equivalent overall system with input $x_c(t)$ and output $y_c(t)$

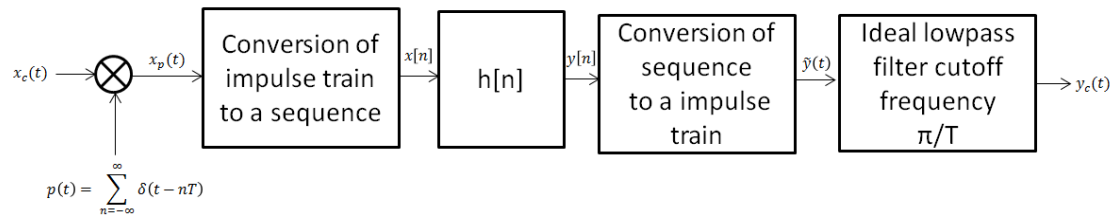


Figure 2

Problem 6 (4'+6'=10')

1) Let $x(t)$ be a real-valued signal for which $X(j\omega) = 0$ when $|\omega| > 2000\pi$. Amplitude modulation is performed to produce the signal

$$g(t) = x(t) \sin \left(2000\pi t + \frac{\pi}{4} \right)$$

A proposed demodulation technique is illustrated in Figure 3 where $g(t)$ is the input, $y(t)$ is the output, and the ideal lowpass filter has cutoff frequency 2000π and passband gain of 2. Determine $y(t)$

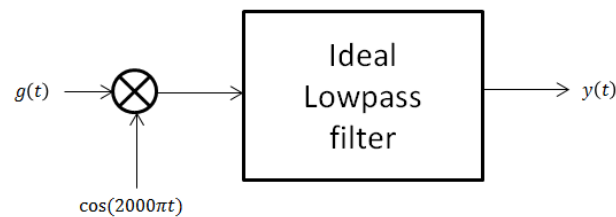


Figure 3

2) Any better design of demodulator? Please explain.